

Seongmin Hong

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My research focuses on the **inversion of deep generative models**, spanning structural design and optimization-based methods. I develop principled algorithms for **exact and approximate inversion** of diffusion models and normalizing flows, with applications to **image editing**, **posterior sampling**, and **inverse problems in imaging**. My doctoral dissertation on this topic was recognized with the **Distinguished Ph.D. Dissertation Award**.

Research Keywords: Diffusion Model Inversion · Normalizing Flows · Latent Diffusion Models · Posterior Sampling · Inverse Problems · Optimization · Image Generation

Education

Ph.D. in Electrical and Computer Engineering – Seoul National University 03 2020 — 08 2025

Dissertation: Inversion of Deep Generative Models: Structural and Optimization-based Approaches

Award: **Distinguished Ph.D. Dissertation**

Advisor: Se Young Chun, GPA: 4.08/4.3

B.S. in Electrical and Computer Engineering – Seoul National University 03 2016 — 02 2020

GPA: 3.70/4.3

Math, Physics, and Astronomy – Gyeonggi Science High School for the Gifted 03 2013 — 02 2016

Experience

Postdoctoral Researcher (Alt. military service) – INMC, SNU 09 2025 — 08 2026

Honors & Fellowships

Seoul Future Foundation AI Seoul Tech Research Support Program, Postdoctoral Fellowship (won 60,000,000 KRW) 2026

Apple Scholars in AI/ML PhD fellowship program - Institutional Nominee 2025

Qualcomm Innovation Fellowship Korea 2024 (won 4,000,000 KRW) 12 2024

Brain Korea 21 Four Program 03 2020 — 08 2025

Top-tier Conference Publications (First-authored)

[DiffBMP: Differentiable Rendering with Bitmap Primitives](#)

Seongmin Hong*, Junghun James Kim*, Daehyeop Kim, Insoo Chung, Se Young Chun (*co-first authored)
Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026.

[Gradient-free Decoder Inversion in Latent Diffusion Models](#)

Seongmin Hong, Suh Yoon Jeon, Kyeonghyun Lee, Ernest K. Ryu, Se Young Chun
Neural Information Processing Systems (**NeurIPS**), 2024.

[Adaptive Selection of Sampling-Reconstruction in Fourier Compressed Sensing](#)

Seongmin Hong, Jaehyeok Bae, Jongho Lee, Se Young Chun
European Conference on Computer Vision (**ECCV**), 2024.

[On Exact Inversion of DPM-Solvers](#)

Seongmin Hong, Kyeonghyun Lee, Suh Yoon Jeon, Hyewon Bae, Se Young Chun
Conference on Computer Vision and Pattern Recognition (**CVPR**), 2024.

[On the Robustness of Normalizing Flows for Inverse Problems in Imaging](#)

Seongmin Hong, Inbum Park, Se Young Chun

International Conference on Computer Vision (**ICCV**), 2023.

[Neural Diffeomorphic Non-uniform B-spline Flows](#)

Seongmin Hong, Se Young Chun

Thirty-Seventh AAAI Conference on Artificial Intelligence (**AAAI**, oral), 2023.

Other Conference Publications

Triadic Dynamics Aware Diffusion Posterior Sampling for Inverse Problems: Optimizing Guidance and Stochasticity Schedules

Junseo Bang*, Dong Ju Mun*, Hoigi Seo, **Seongmin Hong**, Se Young Chun

International Conference on Machine Learning (**ICML**), 2026.

International Journal Publications

[Robust Watermarks for Audio Diffusion Models by Quadrature Amplitude Modulation](#)

Kyoungryeol Lee*, **Seongmin Hong***, Se Young Chun (*co-first authored)

Pattern Recognition Letters, Vol. 198, pp. 22-28, 2025.

[Fidelity-Preserving Zero-shot Diffusion Models for Highly Ill-posed Inverse Problems in Lensless Imaging](#)

Haechang Lee, Dong Ju Mun, Hyunwoo Lee, Kyung Chul Lee, **Seongmin Hong**, Gwanghyun Kim, Seung Ah Lee, Se Young Chun

Image and Vision Computing, Volume 165, January 2026.

[Advanced direction of arrival estimation using step-learned iterative soft-thresholding for frequency-modulated continuous wave multiple-input multiple-output radar](#)

Seongmin Hong, Seong-Cheol Kim, Seongwook Lee

IET Radar, Sonar & Navigation, 2023.

[Radar Signal Decomposition in Steering Vector Space for Multi-Target Classification](#)

Seongmin Hong, Seongwook Lee, Byeong-Ho Lee, Jinwook Kim, Yong-Hwa Kim, Seong-Cheol Kim

IEEE Sensors Journal, 2021.

Preprints & Manuscripts under Review

[DiffStamp: Aesthetically Visible Watermarking by Integrating Differentiable Renderer into Diffusion Models](#)

Seongmin Hong*, Kyungryeol Lee*, Junghun James Kim*, Se Young Chun, 2026

[Principled Reframing in Motion Transfer with Video Diffusion Models](#)

Donggyu Lee*, **Seongmin Hong***, Se Young Chun (*co-first authored), 2025

Invited Talks

[Inversion of Deep Generative Models: Structural and Optimization-based Approaches](#)

At Chung-Ang University, Seoul, Korea, Oct 22, 2025

[Inversion of Deep Generative Models: Structural and Optimization-based Approaches](#)

At ETRI, Daejeon, Korea, Oct 21, 2025

[On the Robustness of Normalizing Flows for Inverse Problems in Imaging](#)

In 2024 SIAM Conference on Imaging Science (**SIAM IS24**), Atlanta, Georgia, U.S., May 28-31, 2024

Exhibitions & Art Projects

Engineer's AI Media Art: Digital Assemblage

02 2026

Led a special workshop at Seoul Science Museum demonstrating DiffBMP technology to create digital assemblage artworks. Taught participants how to use simple recognition methods to transform everyday objects into artistic expressions, exploring the intersection of engineering, technology, and art.

IPA (Interactive Perspective Art): Ciao, Technology

01 2026

Collaborative kinetic art exhibition at SNU Powerplant. Served as the **lead for software infrastructure and typography** for the signature piece *Ordered Volume*—a robotic sculpture utilizing 160 motors and spherical voxels to translate digital data into physical forms.

ARRCO: Multi-view 3D Crystal Point-Cloud Art

2025

Independent crowd-funded project. Developed the full computational pipeline for high-density point-cloud laser engraving to create multi-view optical illusions where distinct images emerge from different observer perspectives.

Patents

Audio Data Management Device and Method Using A Generative AI Model

Se Young Chun, **Seongmin Hong**, Kyungryeol Lee

Korea Patent Filed (Application No. 10-2025-0046656), 04 2025

Teaching Experience

Head TA, Topics in Signal Processing (Generative Diffusion Model, English Lecture)

2025 Autumn

Head TA, Signals and Systems (English Lecture)

2022 Spring

Head TA, Seminar in Electrical and Computer Engineering 3

2020 Spring

TA, Advanced Electromagnetics 1

2020 Spring

Mentoring & Advising

Research Mentor for 10+ Undergraduate Interns

2022 — Present

Intelligent Computational Imaging Lab (ICL), SNU

• **Jaehyeok Bae** (Now Ph. D. student at **Stanford University**) — *ECCV 2024*

• **Inbum Park** (Now Ph. D. student at **University of Maryland**) — *ICCV 2023*

• **Insoo Chung** (Now Research Scientist at **AIRS Medical**)

• **Kyeonghyun Lee** (Now Ph.D. student at **Seoul National University**) — *CVPR 2024*

• **Kyungryeol Lee** (Now Ph.D. student at **Seoul National University**) — *PRL 2025*

• **Dohyun Mah** (Now M.S. student at **Seoul National University**)

• **Donggyu Lee** (Undergraduate Intern) — *Preprint 2025*

• **Mentored 3+ additional interns** on their early-stage research and career development.

Academic Services

Technical Research Personnel (Alternative military service)

03 2023 — 02 2025, 09 2025 — 08 2026

Reviewer: NeurIPS (2024, 2025), ICML (2025, **Gold 2026**), ICLR (2025, 2026), CVPR (2025), ICCV (2025),
AAAI (2026), AISTATS (2025, 2026), ACMMM (2025)
IEEE Sensors Journal, ACM TOMM, IEEE TPAMI, IEEE TMM since 2023
· ICML 2026 Gold Reviewer: awarded to top 25% of reviewers based on area chair ratings

Challenges

[AIM 2024 Depth Upsampling Challenge](#) - 3rd place
Woojae Han, Kyeonghyun Lee, **Seongmin Hong**, Se Young Chun

Workshop Papers

[Unlearning the Unpromptable: Prompt-free Instance Unlearning in Diffusion Models](#)
Kyungryeol Lee*, Kyeonghyun Lee*, **Seongmin Hong***, Byung Hyun Lee, Se Young Chun
Machine Unlearning for Vision (MUV) Workshop in CVPR (extended abstract), 2026.

[Uncertainty-Adaptive Selection of Sampling-Reconstruction for Fourier Compressed Sensing](#)
Seongmin Hong, Jaehyeok Bae, Jongho Lee, Se Young Chun
3rd Workshop on Uncertainty Quantification for Computer Vision in ECCV (extended abstract), 2024.

Domestic Publications

[Optimization of 16-QAM Constellations with AWGN Channel](#)
Seongmin Hong, Jeong-Hoon Park, Doyoung Ham, Seong-Cheol Kim
KICS Summer Conference (in Korean), 2020.

[Target classification based on point cloud data using automotive imaging radar](#)
Sohee Lim, Jaehoon Jung, **Seongmin Hong**, Byeong-ho Lee, Seong-Cheol Kim
KICS Winter Conference (in Korean), 2021.

[Measurement Data Release on 2.4 GHz ISM Band Usage](#)
Doyoung Ham, **Seongmin Hong**, Byonghyo Shim, Seong-Cheol Kim
KICS Winter Conference (in Korean), 2021.

[Analysis of CNN Classifier Performance for Segway Data with 77GHz Automotive Radar](#)
Byeong-ho Lee, **Seongmin Hong**, Jinwook Kim, Seongwook Lee, Byonghyo Sim, Seong-Cheol Kim
KICS Winter Conference (in Korean), 2021.

References

Prof. Se Young Chun (Advisor)
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Prof. Ernest K. Ryu
Assistant Professor, Applied Mathematics
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Prof. Jongho Lee
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